

Question 1

Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain **musicmania.in**.

Steps Performed

1. Downloaded and installed Simple DNS Plus on Windows.
2. Opened Simple DNS Plus.
3. Clicked **New → Primary Forward Zone**.
4. Entered the Zone Name: **musicmania.in**
5. Clicked **Finish** to create the zone.

Result : A new DNS zone named **musicmania.in** was successfully created.

Question 2

Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Steps Performed

- Opened the **musicmania.in** zone.
- Clicked **New A-record**.
- Entered the following details:

Field	Value
Record Name	www.musicmania.in
IP Address	192.168.1.50

- Clicked **OK** to save the record.

Verification (command apply in cmd) :- nslookup www.musicmania.in 127.0.0.1

Output

Server: localhost

Address: 127.0.0.1

Name: www.musicmania.in

Address: 192.168.1.50

Result : The DNS server successfully resolved www.musicmania.in to **192.168.1.50**.

Question 3

Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Steps Performed

- Opened the **musicmania.in** zone.
- Clicked **New CNAME-record**.
- Entered the following details:

Field	Value
Alias Name	play.musicmania.in
Canonical Name	www.musicmania.in

Command Apply in CMD :- nslookup play.musicmania.in 127.0.0.1

Output

Server: localhost

Address: 127.0.0.1

Name: www.musicmania.in

Address: 192.168.1.50

Aliases: play.musicmania.in

Result : The CNAME record successfully redirected **play.musicmania.in** to www.musicmania.in, which resolved to **192.168.1.50**.

Question 4

Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Troubleshooting Steps

Step 1: Verify DNS Zone Configuration

- Open Simple DNS Plus.
- Check whether the zone **musicmania.in** exists.
- Verify that the A record and CNAME record are present.

Step 2: Check Record Details

- Confirm that:
 - www.musicmania.in points to 192.168.1.50
 - play.musicmania.in points to www.musicmania.in

musicmania.in				
Name	Type	TTL	Data	Comments
musicmania.in	NS	10800	diksha	
musicmania.in	SOA	10800	diksha [2026061202]	
play.musicmania.in	CNAME	10800	www.musicmania.in	
www.musicmania.in	A	10800	192.168.1.50	

Step 3: Restart DNS Service

- Save all changes.
- Restart the Simple DNS Plus service.
- Reopen the application.

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C:\Users\cdiks>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\cdiks>nslookup musicmania.in 127.0.0.1
Server:    localhost
Address:   127.0.0.1

Name:      musicmania.in

C:\Users\cdiks>nslookup www.musicmania.in 127.0.0.1
Server:    localhost
Address:   127.0.0.1

Name:      www.musicmania.in
Address:   192.168.1.50

C:\Users\cdiks>ping 192.168.1.50

Pinging 192.168.1.50 with 32 bytes of data:
Reply from 192.168.1.6: Destination host unreachable.
Reply from 192.168.1.6: Destination host unreachable.
Reply from 192.168.1.6: Destination host unreachable.
Reply from 192.168.1.6: Destination host unreachable.

Ping statistics for 192.168.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```